

APPROVED

IT7520: Network and Security Programmability

Course Details	
Course Code:	IT7520
Course Title:	Network and Security Programmability APPROVED
Short Title:	NSP
Course Level::	Level 7
Valid From::	2026/2027 Sem 1
Credits::	15
Owner:	School of ICT
Assessment Method:	Achievement
Course Aim	This course equips students with foundational and advanced skills in network and security programmability using Python. It focuses on the use of Python for socket programming, data encoding, serialization, and automation of network tasks. The course prepares students to analyze, design, and develop secure network applications by integrating modern protocols, APIs, and design patterns. Emphasis is placed on real-world scenarios involving programmability, data communication, and secure application development.

CILO	
On Completion of this course, the learner will be able to	
#	Learning Outcome Description
1	Demonstrate advanced knowledge and understanding of networking and security programmability concepts through Python-based examples and use cases.
2	Analyze data encoding techniques and network protocols to optimize security, efficiency, and interoperability in Python-driven automation.
3	Implement secure network automation solutions using Python frameworks and APIs.
Requisites	
• Pre Requisite: IT6003 • Pre Requisite: IT6008	

Examinations
Assessment Type Examination (Unseen)
Assessment Type Examination (Practical)
No Other Controlled Assessments
Uncontrolled Assessments
Assessment Type Project (Group)

Affiliated Entities			
Entity Code	Entity Title	Entity Version	Entity Type
ICT8030	Bachelor of Information and Communications Technology (Networking Major)	5	Programme
ICT8031	Bachelor of Information and Communications Technology (Networking Major)	4	Programme
ICT8032	Bachelor of Science in Information and Communications Technology (Networking Major)	1	Programme
ICT8050	Bachelor of Information and Communications Technology (Cyber Security Major)	4	Programme
ICT8051	Bachelor of Information and Communications Technology (Cyber Security Major)	4	Programme
ICT8052	Bachelor of Science in Information and Communications Technology (Cybersecurity Major)	1	Programme